

Ahsan Bilal

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RESEARCH INTERESTS

Deep Learning Optimization and LLM Inference: Optimization theory for deep learning and reasoning-oriented LLMs, including post-training methods, preference optimization, alignment, and test-time compute scaling for autoregressive and diffusion language models. Analytical foundations span statistics, probabilistic modelling, and algorithms for scalable NLP and ML systems.

Agentic and Adaptive AI Systems: Design of evolving multi-agent systems with adaptive behaviors, dynamic reputation modelling, and social graph-based coordination. Focus on building scalable, distributed system architectures with strong compliance and connectivity technologies for cross-functional AI product deployment.

Reinforcement Learning for Reasoning and Decision Systems: RL methods for reasoning-oriented LLMs, process reward models (PRMs), and agentic test-time compute allocation, with applications in complex non-stationary environments, signal processing, and ML-driven wireless communication systems (including 5G).

EDUCATION

University of Oklahoma, Oklahoma, USA 2024–Present
Ph.D. in Computer Science; M.S. en route (Deep Learning, Reinforcement Learning, & Reasoning in LLMs)

- CGPA: 3.86/4.0
- Advisor: [Dr. Dean Hougén](#): Research on deep learning optimization, robust LLM reasoning, and RL-based agentic systems.

National University of Sciences & Technology (NUST-SEECS), Islamabad, Pakistan Aug. 2020 – June 2024

Bachelor of Engineering in Electrical Engineering

- CGPA: 3.66/4.0 | Specialization GPA: 4.0/4.0 | Merit Scholarship 2020
- Senior Design Thesis: Person identification using gait with fused graph and 3D-convolutional architectures ([Presentation](#))

RESEARCH EXPERIENCE

REAL Lab, University of Oklahoma Graduate Researcher
Advisor: [Dr. Dean Hougén](#) Aug 2024 – Present

- Research on adaptive test-time compute for LLMs, including verifier/PRM-guided reasoning, agentic RL controllers, trajectory-level search, and dynamic compute allocation across autoregressive (ICML'26) and diffusion language models (COLM'26). Written technical specifications and rigorous statistical analysis for each system; communicated findings cross-functionally across collaborating institutions following collaborative teamwork best practices.
- Work on structured generative models (discrete diffusion), test-time scaling during denoising, and continuous learning diffusion models for wireless channel prediction under distribution shift (ICASSP'26). Maintained well-documented, modular codebases; produced written technical documentation for all research deliverables.

Stanford AI Lab (SAIL), Stanford University Research Collaborator
Supervisors: [Prof. John M. Cioffi](#), [Dr. Emily Fox](#) | Collaborator: [M. A. Mohsin](#) Aug 2024 – Present

- Developed diffusion-based wireless estimation and neural Gaussian radio fields (nGRF), including verifier-guided trajectory search for diffusion test-time scaling and adaptive compute allocation. Applied signal processing techniques and

machine learning (ML) pipelines using pandas and scikit-learn for data preprocessing, classification tasks, and large-scale analysis.

- Worked on continuous learning under distribution shift and multimodal RAG pipelines for wireless systems. Collaborated with stakeholders from Stanford Statistics, Glasgow, and NUST in a cross-functional, collaborative research environment; publications in NeurIPS'25, ICML'25, AAAI'25, ICC'25 (Best Paper), KDD'26.

Conoid.ai (Internet of Agents Platform)

Research / Engineering

Ongoing

- Building a decentralized Internet of Agents platform for self-evolving AI systems, enabling dynamic agent teams based on trust, memory, and prior performance. Designing scalable distributed system architecture and HTTP-based agent communication protocols with security built into all transactions; written system specifications covering all key use cases and acting as a self-starter driving product development across the full engineering flow.

Undergraduate Researcher

Optimal ML Lab, NUST

Supervisor: [Dr. Ahmad Salman](#)

Jan 2023 – June 2024

- Built robust CV/biometric models: boosted-attention ViT for shadow removal and fused GCN + 3D-CNN for gait recognition. Applied analytical methods with attention to detail to achieve state-of-the-art performance on benchmark datasets; supervised experimental pipelines from problem formulation through evaluation.
- Papers under review: Shadow Removal with Boosted Attention, Gait ID using Fused Graph + 3D-CNN.

INDUSTRY EXPERIENCE

Machine Learning Engineer

Islamabad, Pakistan

[Cowlar Design Studio](#) (*Y Combinator 21*) — *Based in USA* Feb 2024 – Aug 2024

- Built an Action Recognition system for Smart Carts (95% accuracy) with dual-inference deployment across edge devices and Nvidia GPU clusters for scalable, real-time inference; developed clean, modular C++/Python components for high-throughput data pipelines.
- Automated fiber-cable alignment via computer vision and ML (96% success), improving precision to $5\mu\text{m}$ and scaling production 40x; reported technical status and findings to cross-functional teams and stakeholders.
- Delivered AI-powered retail automation with Al Meera Consumer Goods Company, Qatar (~QAR 2.9B+ revenue, ~QAR 2.7B market cap), partnering with product and infrastructure teams; added security controls for high-volume transactions and Dash dashboards for monitoring.

SELECTED PUBLICATIONS

Google Scholar

[PC1] What If We Allocate Test-Time Compute Adaptively? [ICML'26](#)

A. Bilal*, A. Mohsin*, M. Umer, A. Subhan, H. Rizwan, A. Mohsin, D.F. Hougen

[PC2] Distilling Disagreement at Test Time Reasoning

[NeurIPS'26](#)

(Submitted)

A. Bilal*, M.A. Mohsin*, M. Umer, E. Fox, D.F. Hougen

[PC3] General Preference Reinforcement Learning [NeurIPS'26 \(Submitted\)](#)

M. Umer, M.A. Mohsin, A. Bilal, E. Fox

* indicates equal contribution.

- [PC4] **Epistemic Uncertainty for Test-Time Discovery** [\(Submitted\)](#) [NeurIPS'26](#)
K. Riaz, M.A. Mohsin, A. Bilal, M. Umer, A. Mohsin, A. Riaz, A. Subhan, J.M. Cioffi
- [PC5] **S^3 : Stratified Scaling Search for Test-Time in Diffusion Language Models** [COLM'26 \(Submitted\)](#)
A. Bilal*, M.A. Mohsin*, M. Umer, D.F. Hougen
- [PC6] **Neural Gaussian Radio Fields for Channel Estimation** [KDD'26](#)
M. Umer*, M.A. Mohsin*, A. Bilal*, J.M. Cioffi
- [PC7] **Pressure, What Pressure? Sycophancy Disentanglement in Language Models via Reward Decomposition** [COLM'26 \(Submitted\)](#)
M.A. Mohsin*, A. Bilal*, M. Umer, E. Fox
- [PC8] **Continuous-Utility Direct Preference Optimization** [\(Submitted\)](#) [EMNLP'26](#)
M.A. Mohsin*, M. Umer*, A. Bilal, Z. He, M.U. Rafique, A. Aali, M.A. Jamshed, J.M. Cioffi, E. Fox
- [PC9] **Refining Over Resampling: Test-Time Self-Correction for LLM Reasoning** [EMNLP'26 \(Submitted\)](#)
A. Bilal, M.A. Mohsin, M. Umer, L. Trigg, A. Subhan, M. Ali, D.F. Hougen
- [PC10] **ITDPDM: Information-Theoretic Discrete Poisson Diffusion Model** [NeurIPS'25](#)
S. Bhattacharya, A.R. Gorle, A. Bilal, C. Ding, A.K.S. Yadav, T. Weissman
- [PC11] **On the Fundamental Limits of LLMs at Scale** [TMLR'26 \(Submitted\)](#)
A. Mohsin, A. Bilal, W. Zhao, M. Umer, Researchers from DeepMind and Meta
- [PC12] **Channel Prediction Under Network Distribution Shift Using Continual Learning-Based Loss Regularization** [ICASSP'26](#)
M.A. Mohsin, M. Umer, A. Bilal, M.I. Qadir, M.A. Jamshed, D.F. Hougen, J.M. Cioffi
- [PC13] **Conditional Prior-Based Non-Stationary Channel Estimation Using Accelerated Diffusion Models** [ICASSP'26](#)
M.A. Mohsin*, A. Bilal*, M. Umer, A. Ali, M.A. Jamshed, D.F. Hougen, J.M. Cioffi
- [PC14] **Transformer-Based Sparse CSI Estimation for Non Stationary Channels** [ICC'26](#)
M.A. Mohsin, M. Umer, A. Bilal, H. Rizwan, S. Bhattacharya, M.A. Jamshed, J.M. Cioffi
- [PC15] **Continual Learning for Wireless Channel Prediction** [ICML'25](#)
M.A. Mohsin, M. Umer, A. Bilal, M.A. Jamshed, J.M. Cioffi
- [PC16] **Task Aware Distributed Source Coding for Correlated Audio Signals Using Perceptual Loss** [AAAI'25](#)
S. Bhattacharya, M.A. Mohsin, A. Bilal, J.M. Cioffi
- [PC17] **Retrieval Augmented Generation with Multi-Modal LLM Framework for Wireless Environments** [ICC'25](#)
M.A. Mohsin, A. Bilal, S. Bhattacharya, J.M. Cioffi
- [PC18] **HDRL for Spectrum Resource Optimization in Integrated Terres-**

trial and Non-Terrestrial Networks [AAAI'25](#)
M.A. Mohsin, H. Rizwan, M. Umer, S. Bhattacharya, A. Bilal, J.M. Cioffi

[PC19] Abstract – LLM for Explainable AI [IEEE DSAA'24](#)
A. Bilal, B. Lin

[PC20] Meta-Thinking in LLMs via Multi-Agent Reinforcement Learning: A Survey [IEEE TAI \(Submitted\)](#)
A. Bilal, M.A. Mohsin, M. Umer, M.A.K. Bangash, M.A. Jamshed

[PC21] On Shadow Removal With Boosted Attention in a Vision Transformer [Springer ML \(Submitted\)](#)
A. Bilal, A. Salman, K. Khurshid, D.F. Hougen

[PC22] Person Identification using Gait with Fused Graph and 3D Convolutional Architectures [ACM TAIS \(Submitted\)](#)
A. Bilal, A. Salman, K. Khurshid

TEACHING EXPERIENCE

Teaching Assistant [University of Oklahoma](#)
[CS-1313: Programming for Non-majors in C](#) *Fall 2024 – Present*

- Supervised weekly lab sessions; designed assignments, graded with clear rubrics, and led help sessions supporting students with C programming. Collaborated with Dr. Neeman to develop supplementary written technical documentation; communicated clearly with students in a collaborative classroom environment to address individual problem areas and foster analytical thinking.

REVIEWER AND TALKS

Conference Reviewer: ICML, NeurIPS, ICASSP, PAKDD.

Journal Reviewer: TMLR, IEEE WCM, Springer MT&A, IP&M, Aquaculture Int., IJIM, IEEE Access, ICES, ISFI.

Talks: Gave a talk on “AI in Healthcare” at Norman Regional Hospital under [Dr. Lubna Mirza](#).

HONORS AND ACHIEVEMENTS

Honorary Certificate of Appreciation: [IEEE Communications Society Student Competition 2025](#) for “Democratizing 6G: AI-Native Wireless Digital Twin for Global Digital Equity and Sustainability.”

Best Student Presentation Runner-up Award: [IEEE DSAA'24](#) Student Forum.

Graduate Fellowship: Awarded Gallogly College of Engineering Graduate Fellowship 2025.

Best Paper Award: [ICC Workshop 2025](#) in Montreal.

Student Travel Grant: [IEEE DSAA 2024](#) in San Diego.

Best Adjudged Industrial Project Award: Final Year Project received [1st place at NUST Open House 2024](#).

UGRIP Selection: Selected as Undergraduate Research Intern for First Cohort by MBZUAI.

IEEE Recognition: Selected as an Emerging Young Researcher in the IEEE Islamabad Section.

Prime Minister Laptop Scheme: Winner of scholarship program.

CERTIFICATIONS

Google UX Professional Certificate

Deep Learning Specialization Certificate

The Advanced Communication Skills Course

SKILLS

Programming: Python, C/C++, Java, Embedded C, MATLAB. Proficient in scripting, algorithm design, and data structures.

Deep Learning & ML: PyTorch, TensorFlow (Keras), OpenCV, Docker, Mlflow, EC2 Instance. Strong machine learning background spanning neural network architectures, natural language processing (NLP), continuous learning, and scalable pipelines for large-scale data processing and dataset management. Proficient in Dash for interactive data analysis and visualization.

Statistics & Theory: Analytical and statistical learning theory, probabilistic modelling, information theory, and optimization methods applied to machine learning research; experienced in rigorous analysis and problem formulation from specification through evaluation.

Infrastructure & Systems: Distributed systems design, HTTP APIs, Docker-based infrastructure with security best practices, scalable deployment on cloud and edge. Familiar with development best practices, including version control, clean code, and written technical documentation; applied rapid prototyping to iterate quickly across diverse use cases.

Web Development: JavaScript (React.js), HTML/CSS, Next.js.

Tools: VS Code, Git, AutoCad, Figma, PyCharm, Raspberry Pi OS, NginX.

Soft Skills: Collaborative teamwork across cross-functional teams; communicate technical status and findings clearly to diverse audiences and stakeholders; supervised junior researchers and student groups; self-starter with attention to detail and innovative problem-solving in fast-paced environments.

Design: Figma, Adobe XD, Adobe Illustrator, Adobe Photoshop, Sketch, WordPress Theme Design.

Core Competencies: Signal Processing, Machine Learning (ML), Classification, Algorithms, 5G, Connectivity Technologies, Compliance, Data Analysis (pandas), Model Development (scikit-learn), Distributed AI Systems.